

Arhitektura računala

LV6

CP – Compare

This instruction performs a compare between two registers Rd and Rr. None of the registers are changed. All conditional branches can be used after this instruction.

(i) $R_d - R_r$

Operands:

Program Counter:

(i) CP R_d, R_r

$$0 \leq d \leq 31, 0 \leq r \leq 31$$
$$PC \leftarrow PC + 1$$

0001	01rd	dddd	rrrr
------	------	------	------

I	T	H	S	V	N	Z	C
—	—	↔	↔	↔	↔	↔	↔

CPI

CPI – Compare with Immediate

Description

This instruction performs a compare between register Rd and a constant. The register is not changed. All conditional branches can be used after this instruction.

Operation:

(i) $R_d - K$

Syntax:

Operands:

Program Counter:

(i) CPI $R_{d,K}$

$$16 \leq d \leq 31, 0 \leq K \leq 255$$
$$PC \leftarrow PC + 1$$

16-bit Opcode:

0011	KKKK	dddd	KKKK
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
-	-	↔	↔	↔	↔	↔	↔

CPSE

CPSE – Compare Skip if Equal

Description

This instruction performs a compare between two registers Rd and Rr, and skips the next instruction if Rd = Rr.

Operation:

- (i) If Rd = Rr then $PC \leftarrow PC + 2$ (or 3) else $PC \leftarrow PC + 1$

Syntax:

- (i) CPSE Rd,Rr

Operands:

$$0 \leq d \leq 31, 0 \leq r \leq 31$$

Program Counter:

$PC \leftarrow PC + 1$, Condition false - no skip

$PC \leftarrow PC + 2$, Skip a one word instruction

$PC \leftarrow PC + 3$, Skip a two word instruction

16-bit Opcode:

0001	00rd	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

SBRC

SBRC – Skip if Bit in Register is Cleared

Description

This instruction tests a single bit in a register and skips the next instruction if the bit is cleared.

Operation:

Operation:

- (i) If $Rr(b) = 0$ then $PC \leftarrow PC + 2$ (or 3) else $PC \leftarrow PC + 1$

Syntax:

Operands:

Program Counter:

- (i) SBRC Rr,b

$0 \leq r \leq 31, 0 \leq b \leq 7$

$PC \leftarrow PC + 1$, Condition false - no skip

$PC \leftarrow PC + 2$, Skip a one word instruction

$PC \leftarrow PC + 3$, Skip a two word instruction

16-bit Opcode:

1111	110r	rrrr	0bbb
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

SBRS

SBRS – Skip if Bit in Register is Set

Description

This instruction tests a single bit in a register and skips the next instruction if the bit is set.

Operation:

- (i) If $Rr(b) = 1$ then $PC \leftarrow PC + 2$ (or 3) else $PC \leftarrow PC + 1$

Syntax:

Operands:

Program Counter:

- (i) SBRS Rr,b

$$0 \leq r \leq 31, 0 \leq b \leq 7$$

$PC \leftarrow PC + 1$, Condition false - no skip

$PC \leftarrow PC + 2$, Skip a one word instruction

$PC \leftarrow PC + 3$, Skip a two word instruction

16-bit Opcode:

1111	111r	rrrr	0bbb
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Zadatak 2.

Program uspoređuje dva prirodna broja koji se nalaze u memoriji na adresi 0x0100 i 0x0101 (potrebno ih prethodno pohraniti u memoriju → preko R4 registra). Zatim ta dva premješta u register R2 i R3 te ih uspoređuje.

- ako je sadržaj registra R2 veći ili jednak od R3 tad se ta dva broja oduzimaju
- ako je sadržaj registra R2 manji od R3 tad se zbrajaju

Rezultat se sprema u memoriju na adresu 0x0102

Rješenje zadatka2

```
.equ broj1= 45
.equ broj2= 125
start:
ldi R16,broj1
mov R4,R16
sts $0100,R4
ldi R16,broj2
mov R4,R16
sts 0x0101,R4
lds R2,0x0100
lds R3,0x0101
```

```
cp R2,R3
BRLO zbrajanje
jmp oduzimanje
zbrajanje:
add R2,R3
sts 0x0102,R2
jmp start
odužimanje:
sub R2,R3
sts 0x0102,R2
jmp start
```

Zadatak 3.

- Napiši program koji u memoriju upisuje 5 prirodnih parnih brojeva počevši od broja 20 (pretpostavka da uvijek bude paran broj). Početna adresa upisa brojeva je 0x0190.
- Nakon upisa 5 brojeva izvršava jednu NOP instrukciju
- Adresiranje memorije se vrši preko Z registra
- Dodatno
 - provjeri najprije dal je početni broj paran ili neparan, te sukladno tome ispisuj brojeve

Rješenje zadatka 3

```
.equ pocbroj=20
.equ broj_brojeva=5
.equ donjidioadrese= 0x90
.equ gornjidioadrese= 0x01
ldi R16,donjidioadrese
ldi R17,gornjidioadrese
mov ZL,R16
mov ZH,R17
ldi R16,pocbroj
ldi R17,broj_brojeva
```

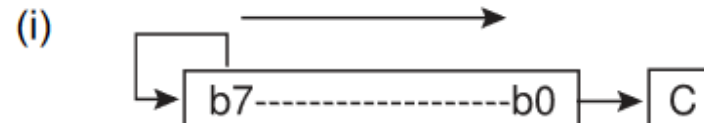
```
start:
inc r16
inc r16
st Z+,R16
dec R17
BRBS 1,kraj
jmp start
kraj:
nop
```

ASR – Arithmetic Shift Right

Description

Shifts all bits in Rd one place to the right. Bit 7 is held constant. Bit 0 is loaded into the C Flag of the SREG. This operation effectively divides a signed value by two without changing its sign. The Carry Flag can be used to round the result.

Operation:



Syntax:

(i) ASR Rd

Operands:

$0 \leq d \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	010d	dddd	0101
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$

Description

Operation:

Diagram illustrating the relationship between a character `C` and a byte array `b7` to `b0`. The character `C` is shown in a box, with an arrow pointing to the start of the byte array. The byte array is represented by a box containing `b7` followed by a dashed line and `b0`. An arrow points from the end of the byte array to the value `0`. A third arrow points from above to the middle of the byte array.

Program Counter:

$$PC \leftarrow PC + 1$$

0000	11dd	dddd	dddd
------	------	------	------

[illegible]

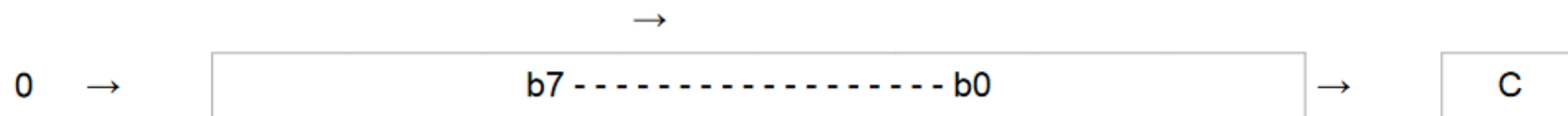
LSR – Logical Shift Right

Description

Shifts all bits in Rd one place to the right. Bit 7 is cleared. Bit 0 is loaded into the C Flag of the SREG. This operation effectively divides an unsigned value by two. The C Flag can be used to round the result.

Operation:

(i)



Syntax:

Operands:

Program Counter:

(i) LSR Rd

$0 \leq d \leq 31$

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	010d	dddd	0110
------	------	------	------

Status Register (SREG) and Boolean Formula

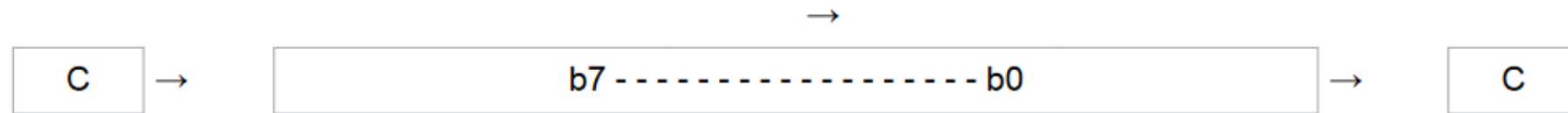
I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	0	$\Leftrightarrow$	$\Leftrightarrow$

ROR – Rotate Right through Carry

Description

Shifts all bits in Rd one place to the right. The C Flag is shifted into bit 7 of Rd. Bit 0 is shifted into the C Flag. This operation, combined with ASR, effectively divides multi-byte signed values by two. Combined with LSR it effectively divides multi-byte unsigned values by two. The Carry Flag can be used to round the result.

Operation:



Syntax:

(i) ROR Rd

Operands:

$0 \leq d \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1001	010d	dddd	0111
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$	$\Leftrightarrow$

Zadatak 1.

- Pomoću navedenih instrukcija (ASR, LSL, LSR, ROL, ROR) izradi proceduru kojom se izvodi instrukcija množenja sa brojem 2.

Zadatak 2.

- Pomoću navedenih instrukcija (ASR, LSL, LSR, ROL, ROR) izradi proceduru u kojoj se zbrajaju bitovi određenog podatka koji je upisan u zadani registar (npr. registra R16)

Zadatak 3.

- Izradi program (proceduru) koji analizira 8 –bitni podatak (broj). Broj dva puta dijeli s dva te mu zaokružuje vrijednost. Vrijednost se zaokružuje na veću vrijednost.

x	x	x	x	x	x	1	1
---	---	---	---	---	---	---	---

Rješenje 3.

Procedura:

LSR R1 ; dijeljenje sa 2

BRCC Div2 ; skače ako nije zaokružen

INC R1 ; zaokruživanje broja

Div2:

LSR R1 ; dijeljenje sa 2

BRCC DivE ; skače ako nije zaokružen

INC R1 ; zaokruživanje broja

DivE: ; izlaz iz procedure